

Domain-specific and domain-general individual differences in visual object recognition

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Abstract

Research in visual object recognition has largely focused on mechanisms common across people but there is increased interest in whether and how people differ in the ability to recognize objects and faces. New tests with a variety of familiar categories are being created and validated to measure domain-specific abilities. Because variability in experience with familiar objects contributes to performance, tests with novel objects were designed and provide evidence for a domain-general visual ability that is relatively independent from general intelligence. These advances have led to improvements in linking activity in some visual areas of the brain with domain-specific experience. Much remains to be done to uncover the neural correlates of domain-general visual ability and assess the predictive ability of visual abilities in real-world settings.

Everyone knows there are important individual differences in cognitive abilities and personality, but in areas without longstanding research traditions, intuitions about individual differences are less clear. Here, I discuss object recognition skills, or the ability to learn to discriminate between visually similar objects, to learn which features are relevant for categorization and individuation, despite image differences such as changes in viewpoint (Palmeri & Gauthier, 2004). How people vary on these abilities has important practical implications, for instance relevant to whether some people may be particularly capable of learning to match fingerprints, to perform radiological diagnosis or to excel at forensic face matching.

Individual differences in cognition or personality deeply influence modern life, which is to some extent accepted because of shared beliefs of large differences in these areas. Therefore, I first sought to measure laypeople's intuitions about visual skills. I asked 100 individuals¹ on Amazon Mechanical Turk to judge how much people might vary on six different tasks, some visual and some non-visual, by rating, from one to seven, how the variability would compare to typical variability in height. I also asked people how they would expect someone who excels at one of these tasks to perform in another. Respondents thought that visual tasks were fairly distinct from other tasks typically related to general intelligence (Figure 1). In addition, respondents judged that people vary less in visual skills than in non-visual skills. Of course, these intuitions may be incorrect and recent studies are beginning to address these and related questions, but it appears that people may be willing to consider that measuring visual skills could add value beyond how we currently compare people.

¹ 56 female, 42 male, 2 unreported; average age = 36.1.

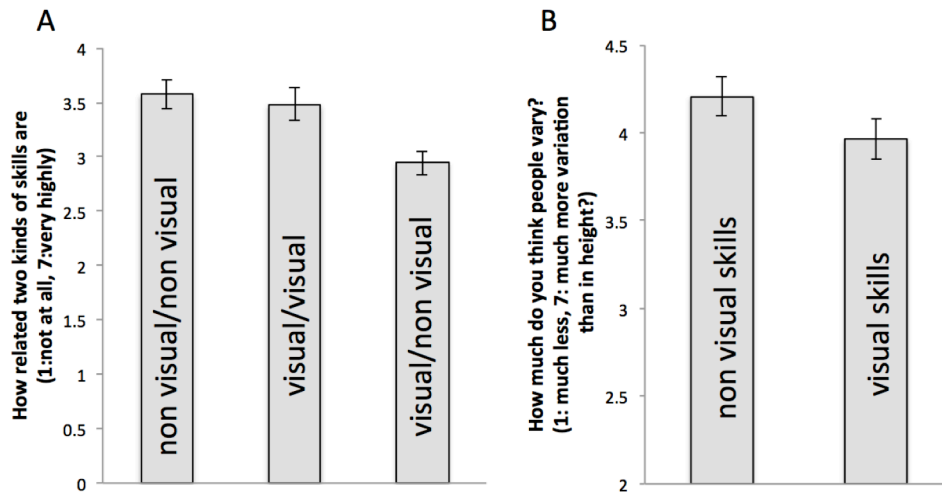


Figure 1. A survey of 100 laypeople suggests that people believe there is a domain-general visual ability that is different from general cognitive skills (A) and which is relatively low in variability (B). Error bars are within-subjects 95% C.I. Non visual skills were learning a second language, new math skills and the produce codes in a grocery store and visual skills were learning to identify cars, to match fingerprints and to identify mushrooms.

CREATING THE TESTS TO STUDY VISUAL ABILITIES

A prerequisite to testing these intuitions is to create psychometrically useful tests. Efforts in high-level vision have only recently focused on creating and evaluating such measures. Although many standard high-level visual tasks are highly sensitive to group-averaged effects (and to some extent for this reason, see Hedge et al., 2017), they have low reliability to measure individual differences (for example, contrast Richler et al, 2014 and Ross et al., 2015, see also Herzmann et al., 2008). Cognitive scientists may sometimes be too quick to label variability across people in their measurements as “individual differences” but the systematic study of individual differences requires first showing that such variability reflects more than measurement error. Duchaine & Nakayama (2006) led the way by creating the first standard test of face recognition ability for use

in the general population, the Cambridge Face Memory Test (CFMT). The CFMT is a learning task, in which people study six unfamiliar male faces and on subsequent trials, select which of three faces is one of these six males. Trial difficulty varies using changing viewpoint and image noise to allow discrimination amongst the best face recognizers. Inspired by this work, my colleagues and I created similar tests for many object categories in the Vanderbilt Expertise Test (VET) battery. Each test requires subjects to study six targets followed by three-alternative forced choice trials (McGugin et al., 2012; VanGulick et al., 2015— see also Dennett et al., 2012). Beyond achieving reliability, scientists must show that their measures have validity. For instance, CFMT performance correlates with other face tasks (Furl et al, 2011; Richler et al., 2011) and separates controls from patients with face recognition deficits (Duchaine & Nakayama, 2006). VET performance for each category correlates with measures of semantic knowledge in the same category, even after controlling for other visual and semantic measures (Van Gulick et al., 2015). Once these basic issues are tackled, experimental researchers expanding into individual differences can benefit from learning new statistical approaches, including factor analysis, structural equation modeling (e.g., Anderson & Gerbing, 1988) and item response theory (Cho et al, 2015; Hambleton et al., 1991).

DOMAIN-GENERAL VISUAL ABILITY IS NOT INTELLIGENCE

Using this first generation of object recognition tests, we can assess whether we truly vary less in object recognition skills than in general intelligence. The coefficient of variation reflects the amount of variation relative to the mean, and being unitless, it allows comparisons of different abilities. For reference, IQ tests have a coefficient of variation of .15 (SD/mean, 15/100). Face recognition, as measured by the CFMT (Duchaine & Nakayama, 2006) in a very large sample

(>44,000 individuals, Germine et al., 2011) had a coefficient of variation of .16 (13/81). In 213 people, six VETs (cars, planes, transformers, shoes, birds, leaves) had a coefficient of variation of .15 and only two tests showed less variation (dinosaurs and mushrooms, .07), possibly because fewer people had a lot of experience with these categories (Van Gulick et al., 2015). Contrary to people's intuitions, evidence suggests that people vary as much in high-level visual skills as they do cognitive skills.

Laypeople also have the intuition that visual skills differ from general cognitive skills. Is this supported by evidence? The bulk of current research suggests that object recognition is independent from other cognitive skills. One of the first studies to suggest that face recognition was relatively independent from cognitive skills (Wilmer et al., 2010) reported that, despite sufficient range and reliability, memory for verbal material only weakly correlated with face recognition ability ($r = .17$). Shakeshaft & Plomin (2015) measured general cognitive ability (g) as a composite of two tests (a verbal ability test and Raven's Progressive Matrices) and found that g showed only a small correlation with face and car recognition ($r = .16$ and $r = .15$ respectively, but see Gignac et al., 2016). Using multivariate genetic analyses in twins, this study also found that while both face and car recognition are highly heritable, this heritability is mostly (>90%) not shared with g . Richler et al. (2017²) measured object recognition for eight categories. On average, object recognition and IQ modestly correlated ($r = .20$) but more importantly regressing out IQ did not reduce the shared variance between ability for one category (say, birds) and the average ability of all seven other categories (cars, dinosaurs, leaves, mushrooms, planes, shoes and transformers on average $r = .47$). Thus, while intelligence might influence performance

² Using a reanalysis of data from VanGulick et al, (2015).

on specific visual ability tests, the underlying domain-general visual ability appears independent from intelligence.

Visual skills may be distinct from intelligence, but the study of visual abilities can benefit from theories in this area. Cattell defined fluid intelligence as the ability to solve new problems and crystallized intelligence as knowledge accumulated from experience (Horn & Cattell, 1966). In recent work we asked whether a domain-general visual recognition mechanism could apply broadly and predict how people acquire new visual skills. Because variable experience with familiar categories can contribute to differences in performance, we created Novel Object Memory Tests (NOMTs, see Figure 2), in a similar format to the CFMT and VET (Richler et al., 2017). We found that performance on these tests varied just as much as on familiar object tests (coefficient of variation between .13 and .15) but showed more shared variance across categories (about 25%) than is typically observed among familiar object tests (about 11%). Importantly, we verified that the ability measured in the NOMTs is not explained by cognitive skills, because shared variance between NOMTs remained unchanged after controlling for performance on the various measures of general intelligence. Researchers attempting to estimate a general visual ability that is distinct from intelligence should consider using NOMTs or other tasks with novel objects to avoid complications from varying experience. Other work (Richler, submitted), shows that tasks with novel objects can predict performance with familiar object categories.

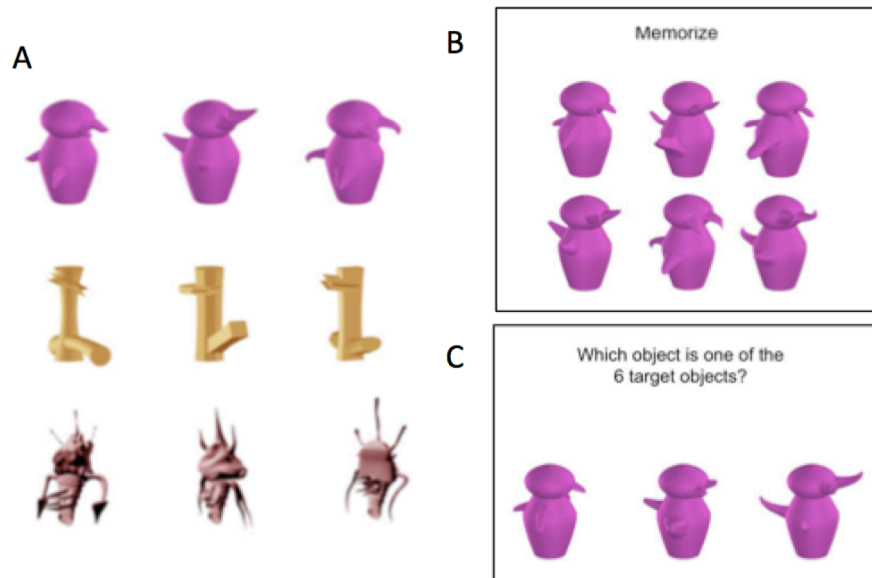


Figure 2. A) Examples of novel objects in NOMTs (Richler et al., 2017). Each object in each set has unique parts although parts can be very similar across objects. There is no rule that can be generalized across categories. B) in each NOMTs, 6 targets are studied individually in two views, then all 6 are shown together to memorize for 20 sec. This is followed by a series of 3-alternatives forced choice trials as shown in C).

Much more work is needed to clarify the nature of the abilities measured in the NOMTs. We have not explored whether the ability they measure is purely visual, and ongoing projects explore discriminant validity against measures of working memory capacity (Xu et al., 2017) and executive functions (Miyake & Friedman, 2012).

CARS (AND FACES) ARE SPECIAL

Interestingly, NOMT performance shows a smaller correlation with face recognition (average $r=.33$) than the correlation between different NOMTs (average $r=.48$). This could lead us to conclude that face recognition is special. However, a second study relates NOMT performance to another category, cars, using the Cambridge Car Memory Test (Dennett et al., 2012, similar to the CFMT and VET-car). Cars are often used to estimate “object recognition ability” in studies using only a single object category (e.g., Shakeshaft & Plomin, 2015). However, in studies that use many familiar categories, performance with cars dissociates from other categories at least as much as faces do (McGugin et al., 2012; Van Gulick et al., 2015). Similarly, NOMT performance was less correlated with cars (average $r = .32$) than different categories NOMTs shared variance (average $r = .48$). Thus, if faces are special because their recognition is poorly predicted by performance with other objects, then so are cars.

Finding that performance with faces and cars is equally independent from performance with other categories (and relatively independent from each other too) was unexpected but important. It offers a new context for studies suggesting that face recognition differs from object recognition based on comparing faces and cars (Dennett et al., 2012, Shakeshaft & Plomin, 2015). A comparison between only two categories (e.g., faces and cars) is not sufficient to claim that one category (faces) is special (Gauthier & Nelson, 2001), because this comparison assumes cars are representative of all non-face object recognition categories. Indeed, when considering a single correlation between two categories (e.g., $r = .37$ between face recognition and car recognition, Dennett et al., 2012), one can emphasize either common variance that suggests overlapping

mechanisms or the unshared variance (unrelated to measurement error) suggesting independent mechanisms. For perspective, the mean pairwise correlation between any two non-face object recognition tests (e.g., birds, cars, planes, dinosaurs, shoes) are about the same magnitude ($r = .33$; McGugin, Richler, Herzmann, Speegle, & Gauthier, 2012; Van Gulick, McGugin, & Gauthier, 2015). Importantly, these pairwise correlations vary greatly, ($r = .0-.5$), and performance for faces and cars, across different datasets, consistently show the lowest correlations with performance for other categories (McGugin, Richler, Herzmann, Speegle, & Gauthier, 2012; Van Gulick, McGugin, & Gauthier, 2015, Richler et al., submitted).

If only faces are special, an evolutionary account, while difficult to test, at least seems plausible. In contrast, car recognition may seem special for multiple reasons: people vary more in their car experience than other categories, cars are more familiar to most people and we have more semantic information about cars than many categories. So far, we find little support for most of these hypotheses. No measure of car experience or car knowledge mediated the relation between car and other category performances (Richler et al., in press). However, measuring experience with a category is particularly challenging (Gauthier et al., 2014) and we only measured knowledge of car model names as a proxy for semantic car knowledge (Van Gulick et al., 2015). One interesting way to measure experience is population density: one report found that face recognition increases with increases in hometown population density (Balas & Saville, 2015). While population density may not influence experience for all categories, it would also apply to cars. Perhaps with increasing experience people rely on different recognition mechanisms for faces and cars, reducing the contribution of domain-general variance in these cases. Of course, car and face recognition do not have to be “special” for the same reason(s), but until we

understand why car recognition is relatively distinct, we cannot rule out that the same explanation can account for how both face and car recognition become special. After all, both car and face recognition show a large and equivalent degree of heritability (Shakeshaft & Plomin, 2015, Wilmer 2010), and the car recognition ability repeatedly correlates with neural responses to cars in the fusiform face area, FFA (Gauthier et al., 2000, Xu, 2005, McGugin et al., 2012b; 2015). These results certainly raise new questions in a field often motivated by differences between face and non-face object recognition (e.g., Farah et al., 1998, Kanwisher, 2000).

MEASURING DOMAIN-SPECIFIC VISUAL ABILITIES

In the first studies looking at correlates of real-world expertise (Gauthier et al., 2000; Xu et al., 2005), performance for a category of interest (say, cars) was compared to a baseline for another single category (birds), and the difference in performance was related to the difference in neural response for cars-minus-birds, and vice versa (in the same subjects birds-minus-cars was used to quantify bird expertise and related to the birds-minus-cars neural response. While this led to replicable expertise effects for objects in FFA, there are limitations to this approach. First, when using this subtraction approach, the control category variance can contribute as much variance as the main category, so a car expertise score may be a “car expertise relative to bird expertise” score. A more appropriate approach may be to regress out the control category variance (DeGutis, et al., 2013). Second, even using regression, it is unlikely that a single control category could accurately represent “generic” object recognition mechanisms. Third, using only a single task format means that some individual differences could be task-specific rather than general. Thus, in recent studies of expertise we use several control categories (e.g., comparing cars to seven other categories) and several tasks (e.g., matching and learning tasks, McGugin et al.,

2015). A definition of domain-specific expertise that transcends task and control category can increase validity of measurements and should lead to predictions of behavior and neural effects in a wider range of situations.

FUTURE DIRECTIONS

The work I discussed is a stepping-stone to several new avenues of research. For instance, past studies of expertise have uncovered neural substrates related to performance with a single category like cars and as such point to the neural substrates of category-specific effects likely related to experience. New studies focusing on domain-general visual skills are needed to reveal aspects of brain structure and activity support the ability to acquire expertise in new categories. As we learn more about domain-general visual abilities, it will be important to assess their predictive validity in real world domains such as medical imaging. As we take on this challenge, it is important to remember that the quality of our predictors cannot make up for poor measurements of the criteria we want to predict. Additional work may be required to achieve reliable measures of the quality of decisions in professional areas. In addition, experts are by definition rare, limiting sample sizes, and more longitudinal efforts are needed to study the trajectories from novice to expert, which may be constrained by visual abilities (Gegenfurtner et al., 2017).

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Recommended readings.

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2. Duchaine, B., & Nakayama, K. (2006). The Cambridge Face Memory Test: Results for neurologically intact individuals and an investigation of its validity using inverted face stimuli and prosopagnosic participants. *Neuropsychologia*, 44(4), 576-585. A landmark study in the creation of tests to study visual abilities in the normal population.
3. Richler, J.J., Wilmer, J.B., Gauthier, I. (2017). General object recognition is specific: evidence from novel and familiar objects. *Cognition*, 166: 42-55. A recent study offering the first tests designed to measure domain-general object recognition.
4. Shakeshaft, N. G., & Plomin, R. (2015). Genetic specificity of face recognition. *Proceedings of the National Academy of Sciences*, 112(41), 12887-12892. A twin study showing that face and car recognition are heritable and independent from intelligence.

5. Van Gulick, A.E., McGugin, R.W. & Gauthier, I. (2015). Measuring non-visual knowledge about object categories: The Semantic Vanderbilt Expertise Test, *Behavioral Research Methods*, 1-19. A study offering a battery of tests for visual abilities and semantic knowledge for several object categories that can be used in future work.

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